8. NETWORK TROUBLESHOOTING, MANAGEMENT, MONITORING AND OPTIMIZATION

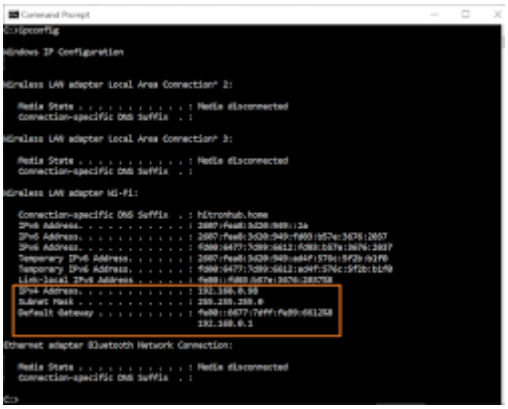
Course: Diploma in Cyber Security

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Introduction

Network management means checking the network regularly, finding problems and improving performance. Monitoring helps identify issues before they become serious.

Task 1: ipconfig /all



Task 2: Disconnect/reconnect test

Disconnect Wi-Fi/Ethernet and ping www.google.com. Reconnect and repeat. Compare the result and explain what timeout or successful replies mean.

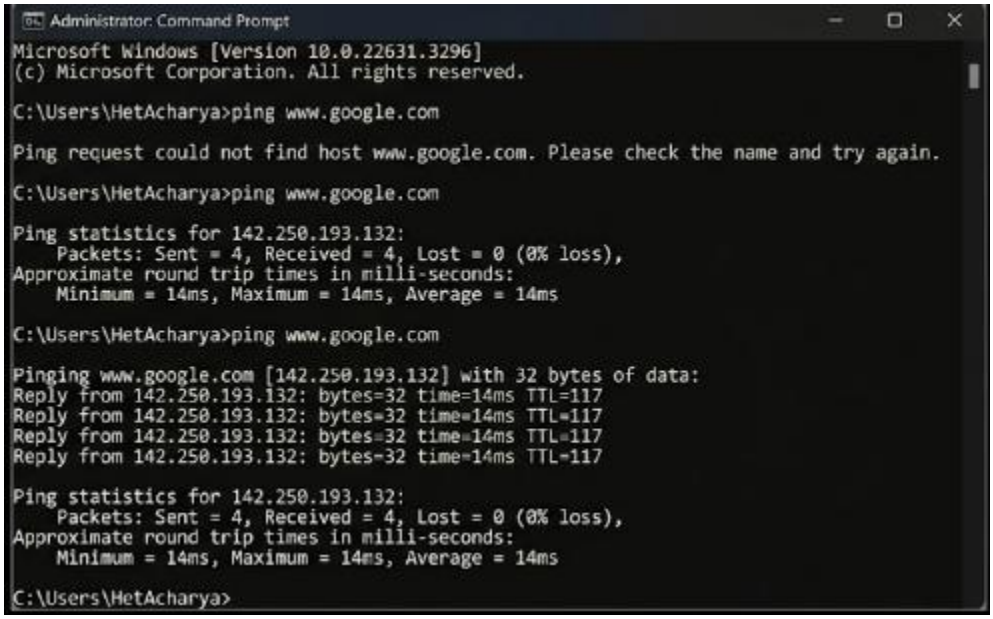


Figure 2: ICMP Echo Failure During Interface Disconnection vs Successful Post-Reconnection Recovery

Task 3: Monitor network usage

Open Task Manager or Resource Monitor and monitor network use while streaming YouTube for about five minutes. Record the peak value and process using the most bandwidth.

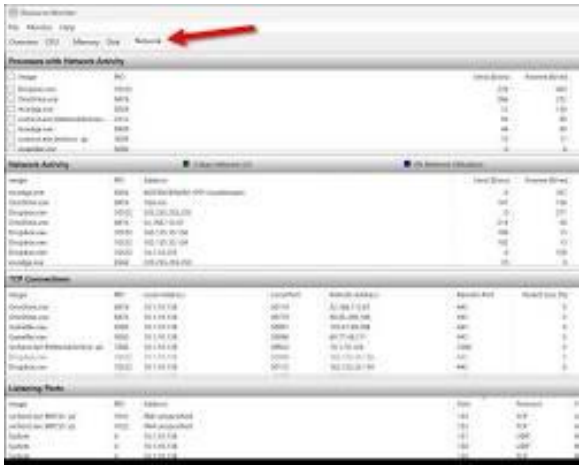


Figure 3: Windows Resource Monitor Real-Time Network Bandwidth Spike During Media Streaming

Task 4: Wireshark filter

Capture at least 30 seconds of traffic while browsing a permitted website. Use the filter bar and inspect HTTP/HTTPS-related traffic. List three types of information visible at the packet level, without collecting private data.

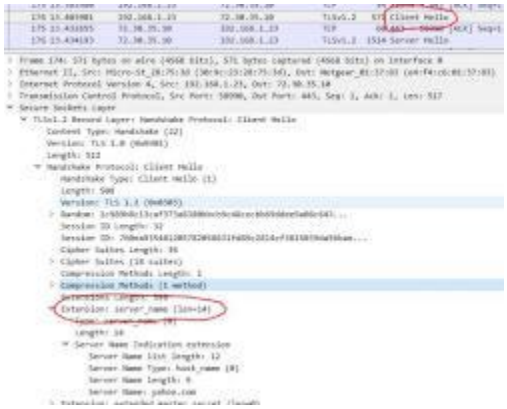


Figure 4: Wireshark Display Filter Inspection Isolating TLS Handshake and TCP Session Headers

Task 5: Optimization checklist

A useful checklist can include checking Wi-Fi signal, reducing interference, updating drivers, using Ethernet for gaming where possible, restarting network equipment, and checking background downloads.

Optimization Domain	Action Item & Implementation Step	Expected Technical Benefit
Physical Media	Deploy shielded Cat6 Ethernet cabling directly for latency-sensitive workloads (gaming/servers).	Eliminates Wi-Fi packet drop, jitter, and radio frequency interference; reduces ping to < 1 ms.
Wi-Fi Frequency Bands	Migrate high-bandwidth devices from congested 2.4 GHz channels to 5 GHz or 6 GHz (Wi-Fi 6E).	Increases channel width (up to 80/160 MHz) and eliminates co-channel microwave/Bluetooth interference.
Radio Channel Selection	Use a Wi-Fi analyzer to manually assign non-overlapping channels (1, 6, or 11 on 2.4 GHz).	Prevents packet collisions with adjacent neighboring wireless access points.
Network Interface Card	Regularly update wireless NIC and LAN adapter chipset drivers from OEM vendor portals.	Resolves known OS kernel packet handling bugs, sleep-state drops, and connection

Optimization Domain	Action Item & Implementation Step	Expected Technical Benefit
		negotiation flaws.
Gateway Maintenance	Schedule routine power-cycles (reboots) of residential modem/router gateways.	Clears bloated NAT state tables, resolves memory leak degradation, and forces channel re-negotiation.
Background Optimization	Configure Windows Delivery Optimization and cloud storage apps (OneDrive/Drive) to throttle background uploads.	Prevents bufferbloat and latency spikes caused by asymmetric upstram saturation.

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.